Edge-Guided Adjacent Scale Fusion Network for Building Change Detection With VHR Remote Sensing Images

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***Abstract*—Building change detection (BCD) with very high-resolution (VHR) remotely sensed images (RSIs) is crucial in various remote sensing applications. However, most existing multiscale methods mainly aggregate features from different levels, while the relationships and discriminative differences between adjacent-scale features are seldom explicitly modeled. Existing edge-guided methods generally use building boundaries only as auxiliary constraints, without explicitly exploiting building-edge differences between bitemporal images. In this work, motivated by the idea of utilizing multiscale feature difference, a novel encoder–decoder network called edge-guided adjacent-scale fusion (EGASF) network is proposed for BCD with VHR RSIs. First, taking into account the connection of a target between different scales, an adjacent-scale feature fusion (ASFF) module is constructed to join the features from different scales. Second, a multiscale feature difference metric (MFDM) module is proposed to measure the change magnitude between the learned features of different scales. Finally, a building-edge difference extraction (BEDE) module is proposed and coupled with the learned multiscale difference feature to guide the decoding progress for the task of BCD, complementing the loss of information at the stage of the encoder. Experiments on four pairs of real BCD datasets effectively verified the superiority of the proposed EGASF network compared with eight state-of-the-art methods. Compared with the best-performing competing method on each dataset, the proposed EGASF achieves absolute OA improvements of 0.30, 0.82, and 0.52 percentage points on Data-A, Data-B, and Data-C, respectively, while obtaining a comparable result on Data-D. The code for the proposed EGASF network can be available at https://github.com/ImgSciGroup/EGASF-Net**

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***Index Terms*—Building change detection (BCD), deep learning network, very high-resolution (VHR) remote sensed images.**

I. Introduction

BUILDINGS are one of the important artificial structures on the Earth's surface. Monitoring changes in buildings in a geographic area is crucial for conducting relief efforts after an Earthquake in a city area [1], making urban development plans [2], analyzing land cover and land use [3], calculating the urban growth direction [4], etc. In recent decades, the rapid advancements in satellite and aerial technology have provided an opportunity to obtain remotely sensed images (RSIs) with very high spatial resolution. Comparison of bitemporal very high-resolution (VHR) RSIs is a convenient method for achieving the task of building change detection (BCD). However, BCD with VHR RSIs remains challenging because repeated feature downsampling may cause the loss of fine building details, buildings exhibit considerable scale variations, and complex boundaries and imaging differences can introduce pseudo-changes.

Various BCD approaches have been promoted and successfully applied in practical applications [1]. Herein, these approaches can be broadly categorized into two categories, namely, traditional methods (nondeep learning-based methods) [5] and deep learning methods [6], [7], [8].

Traditional BCD with RSIs primarily focuses on improving BCD performance by exploring the spatial context features surrounding a central pixel, such as morphological building index [9], adaptive local textural feature extraction [10], and Markov random field-based methods [11]. Meanwhile, object-based change detection methods are effective for utilizing neighborhood information [12]. Finally, some general CD methods, such as CVA-FCM [13], SVM [14], and MLS [15], can also be used for BCD. Although these approaches based on traditional methods can obtain satisfactory BCD maps with RSIs, the performance of some approaches depends on the empirical parameter's setting.

In addition to traditional methods, deep learning-based BCD can provide excellent application performance and has been widely used in the community of BCD with RSIs, owing to the deep and abstract feature learning ability of deep learning networks [16]. U-Net is a classical network used in BCD with RSIs [17]. This network was first proposed for

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biomedical image segmentation [18], and its architecture is derived from the "fully convolutional network." Accordingly, various BCD deep learning approaches based on the backbone of U-Net are proposed, including U-Net with multiscale feature fusion [19], combined with attention mechanism [20], [21] or transformer mechanism of U-Net [19], [20], [22], etc. The U-Net allows the network to simultaneously explore global and contextual information and efficiently extract and utilize multiscale features through skip connections to obtain enhanced performance [23]. However, the U-Net suffers from the loss of structural information during the extraction of multiscale features and cannot fully use the relationships between adjacent-scale features.

Meanwhile, differential feature fusion is another way to improve the detection performance of BCD with RSIs. In the BCD task, during the process of extracting building features, the network will also extract some nonbuilding features. These nontarget features affect the improvement of detection performance. Accordingly, we need to pay attention to the temporal correlation between bitemporal images to suppress the interference of pseudo-changes. The differential feature fusion strategy extracts diverse features by expanding the distance between changed features and narrowing the variation between unchanged features. Bai et al. [24] used long short-term memory (LSTM) to mine the temporal dependence of bitemporal features to obtain differential information. Chen et al. [25] utilized channel attention mechanisms (CAMs) to enhance interclass differences. Luo et al. [26] proposed a bidirectional diff-changed feature representation module to learn the fine change features of bitemporal images at different scales to enhance the discriminative performance of subtle changes. These methods avoid feature redundancy caused by directly concatenating bitemporal image feature maps. Instead, these methods selectively transport interesting features to the network through differential feature fusion. How to construct a suitable and effective differential feature fusion mechanism is an important problem in the BCD task.

Aside from difference feature fusion approaches, making full utilization of edge prior knowledge is a way to improve the detection of building change states. Boundary information for buildings can reflect not only their geometric characteristics but also their location information [27]. Consequently, various edge-guided methods, such as edge neighborhood contrastive learning for BCD [28], edge-guided recurrent convolutional neural network [24], edge-guided parallel network [29], BCD with edge guidance and differential enhancement network [30], and multitask difference-enhanced Siamese network for BCD in RSIs [31], have been promoted in practical applications owing to the traits of boundary information. However, these methods simply use prior edge knowledge to constrain the model to learn edge features and do not combine edge features with the task of change detection. The recognition of boundary features is often insufficient due to the mutual interference between the background and the building features near the edge. Compared with traditional BCD methods, deep learning-based approaches can learn more discriminative semantic representations but may lose fine structural information during repeated downsampling. Existing multiscale and difference-based methods mainly aggregate features across different levels or compare bitemporal features, while the relationships and discriminative differences between adjacent-scale features remain insufficiently explored. Moreover, edge-guided methods often use boundary information only as auxiliary supervision or refinement cues. In contrast, the proposed EGASF network jointly models adjacent-scale feature relationships, multiscale feature differences between bitemporal images, and building-edge differences within a unified encoder–decoder framework.

Despite their effectiveness, existing BCD methods still exhibit three main limitations, including structural information loss during encoding, insufficient modeling of adjacent-scale feature relationships and differences, and inadequate exploitation of building-edge differences between bitemporal images. These limitations are detailed as follows.

1) Information loss is inevitable in the image feature encoding progress of a deep learning network. Intuitively, a deep learning network can encode an input feature through a resampling operation to obtain deep and abstract features. However, detailed information may be lost in the stage of the encoder. How to reduce information loss when encoding the features of buildings is challenging for BCD with RSIs.
2) The same target is typically presented with different shapes and sizes in various scale observations. The convolutional features from different convolution layers produce varying outcomes, and numerous existing approaches overlook these differences in the decoding progress. How to utilize the multiscale feature difference from each convolution layer in an encoder to enhance the performance of the decoder remains challenging.
3) Various studies indicated that the edge of buildings plays an important role in the task of BCD with RSIs. Nevertheless, the method of quantifying the edge change between bitemporal images remains a challenge. The same buildings may perform with different edge detection results due to the varying imaging conditions of bitemporal RSIs. Consequently, distinguishing the similarity between the buildings and utilizing the building-edge-change difference to improve the performance of BCD presents a significant challenge.

In this work, a novel encoder–decoder network called adjacent-scale feature fusion (ASFF) and edge-guidance (EGASF) network is proposed for BCD with bitemporal VHR RSIs. The major contributions of the work are summarized as follows.

1) An EGASF network was first proposed in the community of BCD with RSIs. The proposed network concentrated on exploring the multiscale and building-edge feature differences to guide the decoding progress and improve the performance of BCD with RSIs. To the best of our knowledge, this work is the first to focus on the feature difference among a pair of two adjacent layers. Moreover, multiscale and building-edge feature differences are proposed and coupled for improving the performance of BCD.

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2) An ASFF model is proposed to capture the feature connection between two adjacent layers. An inputting feature was processed by a series of convolutional layers. Accordingly, a connection exists between two adjacent layers. Here, an ASFF model is proposed to capture the connection between the features from two adjacent layers to guide the decoding for BCD with RSIs.
3) The multiscale feature difference metrics (MFDMs) model and the building-edge difference extraction (BEDE) module are first proposed to guide the decoding progress for improving the performance of BCD with RSIs based on the connection feature extraction via the ASFF model.

The remainder of this article is organized as follows. Section I introduces the related work. Section III presents the details of the proposed approach. Section IV provides the specific experimental deployment. Finally, Section V presents the conclusion.

II. Related Works

A. Encoder–Decoder Network

An encoder–decoder structure has been devised for semantic segmentation in natural image processing [32]. Given that an encoder–decoder structure can fully explore the context features between images, it is also widely used in CD with RSIs [33]. In recent years, numerous researchers have utilized encoder–decoder models based on different baseline networks to solve CD tasks. For example, an encoder and decoder network is combined with ResNet [34], dense connected Siamese network-based U-Net [35], and RNN-based encoding and decoding network [36]. Moreover, an attention mechanism is introduced to address the unavoidable problem of loss of detailed information during sampling. This mechanism allows the model to focus on changing areas in RSIs [37]. Zhang et al. [38] proposed an attention-guided edge refinement network for BCD. Zhang et al. [38] combined densely connected convolution block with channel attention and proposed a densely connected Siamese network based on an encoder–decoder structure. However, the attention approaches failed to adequately model the complex scenarios and rich semantic information of RSIs. Accordingly, some researchers have introduced transformers in networks based on encoder–decoder structures to model contexts in the spatial–temporal domain, such as BIT [39] and PA-Former. In summary, the encoder–decoder structure can be easily combined with other mechanisms to extract features. Therefore, we use the encoder–decoder structure as the architecture of the proposed network to extract deep features of buildings.

B. Multiscale Feature Extraction

Since the feature pyramid network (FPN) was proposed, multiscale feature extraction methods have been widely used in various computer vision [40]. Given that the FPN can capture and combine the features of different scales to mine high-level features, which helps the network better focus on the target, numerous researchers have applied multiscale learning in change detection tasks with RSIs. Various CD multiscale feature extraction methods are proposed. These methods include using the Siamese network with temporal attention modules as a feature encoder to extract multiscale features from bitemporal RSIs [23]. Another approach involves utilizing a CNN-transformer network with multiscale context aggregation for efficient farmland change detection [41]. Additionally, another method uses the local–global pyramid structure to expand the reception field to obtain robust features of different objects with multiple scales [19]. Furthermore, a multiscale decoupled convolution design has been suggested to flexibly capture the multiscale features of changed objects [21]. Existing multiscale methods mainly differ in their feature construction and aggregation strategies, including temporal attention, local–global pyramids, multiscale convolutions, and change-gradient guidance. Nevertheless, these methods mainly focus on multiscale feature extraction or fusion, while the correspondence and discriminative differences between adjacent-scale features are rarely explicitly modeled. Lv et al. [42] proposed a multiscale attention network guided with a change-gradient image for land-cover change detection. In view of the advantages of multiscale feature extraction, in the proposed network, we use an encoder–decoder structure to obtain multiscale features and use the ASFF module to fuse features at different scales to generate enhanced contextual multiscale feature representations while guiding the network's detection accuracy of building changes.

Recent advances in intelligent remote sensing analysis have increasingly focused on large-scale pretraining and foundation models. Hong et al. [43] systematically summarized hyperspectral imaging principles, processing workflows, and emerging AI-based analysis methods. SpectralGPT introduced a spectral remote sensing foundation model based on 3-D generative pretraining and demonstrated its transferability to multiple downstream tasks [44]. FlexiMo further improved the adaptability of remote sensing foundation models to varying spatial resolutions and channel configurations [45]. Different from these general-purpose pretrained models, the proposed EGASF is specifically designed for bitemporal VHR BCD and explicitly models adjacent-scale feature relationships and building-edge differences.

III. Method

A. Overview

The proposed EGASF network is mainly composed of encoding and decoding parts to achieve the BCD task with RSIs (Fig. 1). The encoding part of the EGASF network is divided into two branches. Each branch has five convolution layers. The decoding part contains five convolution layers. In the proposed EGASF network, three submodules, including ASFF, MFDM, and BEDE modules, are proposed and embedded in the backbone of the proposed network. Given the bitemporal images, the two encoder branches first extract hierarchical features at five scales. The ASFF module models the relationships between adjacent-scale features in each branch, and the MFDM module subsequently measures the differences between the corresponding bitemporal features. The BEDE module further enhances the building-edge difference information. Finally, the resulting features are delivered

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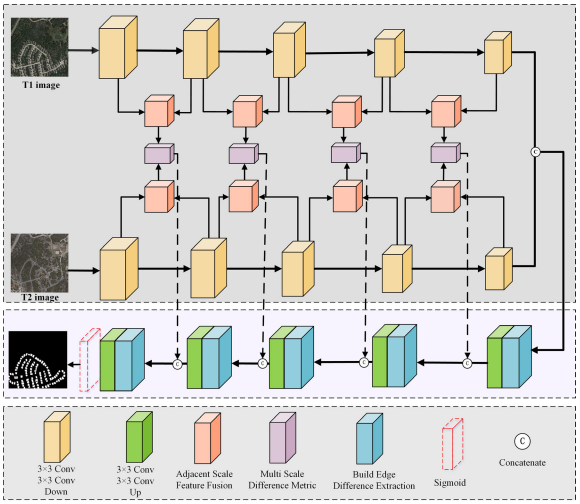


Fig. 1. Flowchart of the proposed EGASF-based network for BCD with bitemporal RSIs.

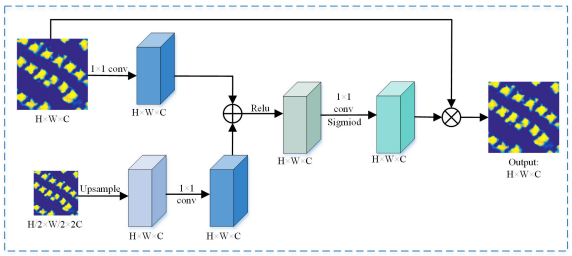


Fig. 2. Flowchart of the ASFF module.

to the corresponding decoder layers to progressively generate the building change map.

B. ASFF Module

Various studies indicated that exploring multiscale features can enhance the performance of change detection with RSIs. However, the connection relationship between different convolution layers is neglected. In this section, the ASFF module was proposed to explore the feature connection from different layers. The structure of the ASFF module contained two branches, which are fed by the features (X and Y) from two adjacent layers (Fig. 2). Here, $X \in R^{H\times W\times C}$ and $Y \in R^{H/2\times W/2\times 2C}$ are the two features from two adjacent layers. In this work, feature consistency refers to the correspondence between the spatial and channel responses of X and Y' for the same target. A larger value in F indicates a response jointly supported by both adjacent scales, whereas a smaller value indicates a weak or inconsistent response. Finally, the consistency of the input features (X and Y) is calculated, which can be expressed as F, as shown in the following equation:

$$F = \sigma\left(Conv\left(R\left(Conv1\left(X\right) \oplus Conv1\left(Y'\right)\right)\right)\right) \tag{1}$$

where $\sigma(\cdot)$ represents a sigmoid activation function, $R(\cdot)$ represents a ReLu activation function, $Conv1(\cdot)$ represents a 1×1 convolution operation, and $\oplus$ represents elementwise summation. Thereafter, the consistency enhancement feature $F \in R^{H\times W\times C}$ and the initial feature map (X) are fused through

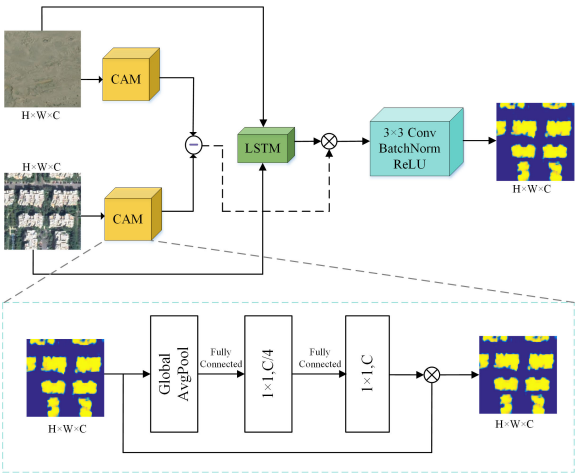


Fig. 3. Structure of the MFDM module.

elementwise multiplication to obtain the enhanced multiscale $X' \in R^{H\times W\times C}$ (see the following equation):

$$X' = F \otimes X \tag{2}$$

where $\otimes$ represents elementwise multiplication. The operations in the proposed ASFF module demonstrated that the enhanced feature map X' fused the multiscale feature from different convolutional layers, thereby highlighting the connection between the pairwise of the adjacent feature.

C. MFDM Module

After the connection features are obtained from adjacent layers using the ASFF module, their change magnitudes between bitemporal images still need to be explicitly measured. Therefore, the MFDM module is designed to quantify and enhance these feature differences. A CAM is first embedded in the MFDM module to highlight the interesting areas (Fig. 3). The inputting features are denoted as $X1 \in R^{H\times W\times C}$ and $X2 \in R^{H\times W\times C}$. Thereafter, the attention maps based on X1 and X2 are subtracted to obtain the building change attention map. In addition, an LSTM network [46] is suggested to measure the similarity between the pairwise inputting features (X1 and X2) to accurately extract the changed buildings in the difference feature map. Here, the learned similarity from the pairwise inputting features via LSTM is taken as a weight for the subsequent operation. If the similarity is higher, then the weight is larger. By contrast, a smaller similarity means a lighter weight. The enhanced difference feature map $X' \in R^{H\times W\times C}$ can be formulated follows the following equation:

$$X' = |AM\left(X1\right) - AM\left(X2\right)| \otimes L\left(X1, X2\right) \tag{3}$$

where $AM(\cdot)$ represents a channel attention operation, $L(\cdot)$ represents the LSTM that measures the similarity of pairwise features, and $\otimes$ is elementwise multiplication.

Finally, a convolution with batch Norm ReLu was provided to smoothen the salt-and-pepper noise in X' to obtain the final multiscale feature difference map (X''). Based on the provided clarification, the proposed MFDM can focus on the interesting area, and the changed area can be enhanced.

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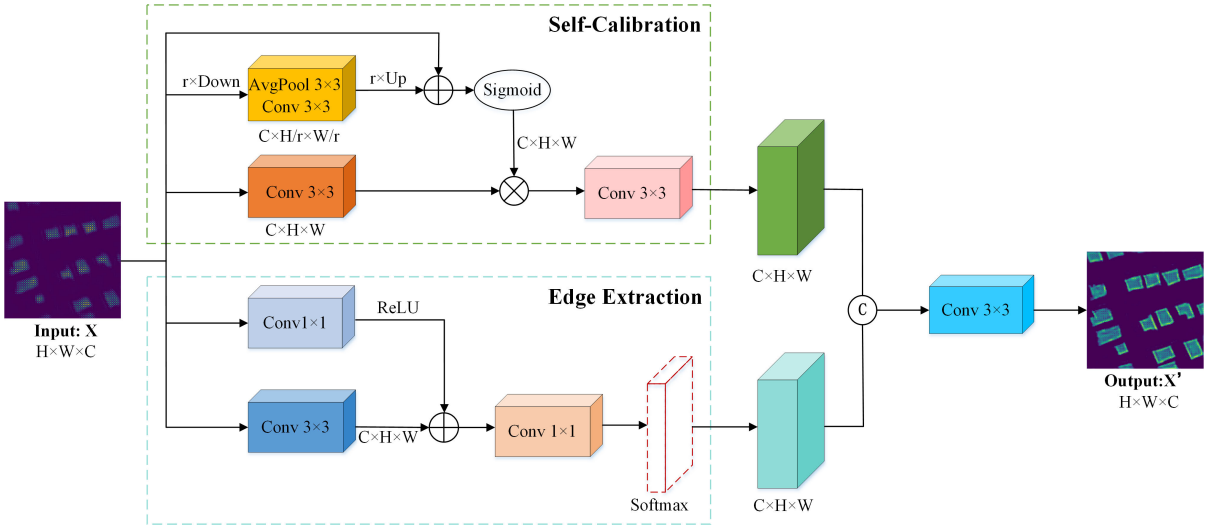


Fig. 4. Proposed structure of the BEDE module.

D. BEDE Module

The edges of buildings play an important role in recognizing the building change. Accordingly, the BEDE module is proposed to concentrate on detecting the edge difference. The proposed BEDE module consists of two branches (Fig. 4). However, building-edge differences between bitemporal images remain insufficiently exploited. Therefore, the BEDE module is designed to extract these differences and enhance building-boundary information. Given an input feature X_1, the self-calibration branch enhances spatially coherent building change features, while the edge-extraction branch emphasizes local boundary responses. In the first branch, a well-known self-calibrated convolution design [47] is introduced here to enhance the representation learning capacity of the traditional CNNs. Various studies have indicated that the self-calibrated convolutions are simple and easy to be applied to augment standard convolutional layers without adding the complexity and parameters of the backbone. In the self-calibrated convolution design, the inputting feature is presented as X_1, and the average pooling with filter size 3×3 convolution is designed to obtain spatially varying features W_1 as follows:

$$W_1 = conv\left(AvgPool\left(X_1\right)\right). \tag{4}$$

Subsequently, W_1 is upsampled to the same spatial resolution as X_1, allowing the contextual feature to be fused with the original local feature. Thereafter, a fusion of spatially varying features with initial features is obtained as follows:

$$W_1' = conv\left(conv\left(X_1\right) \otimes \sigma\left(X_1 \oplus W_1\right)\right) \tag{5}$$

where $conv(\cdot)$ represents a 3×3 convolution operation, $\sigma(\cdot)$ represents a sigmoid activation function, $\oplus$ indicates elementwise summation, and $\otimes$ denotes elementwise multiplication.

The second branch is about the edge-guidance module [24] because the other branch extracts the building-edge information and estimates the change information through 1×1 convolution block. Furthermore, an edge-changed probability map is obtained through the softmax layer. The edge-extraction branch applies a 1×1 convolution for channel projection and a 3×3 convolution for capturing local boundary responses. Their outputs are summed and further processed by a 1×1 convolution, followed by a Softmax operation to generate the edge-change probability map. Finally, the results of the two branches are concatenated, and a 3×3 convolution is utilized to obtain the final change probability map $X' \in R^{H \times W \times C}$ with edge enhancement information.

E. Loss Function

In the training progress, the BCD task can be regarded as a binary classification task. Accordingly, a binary cross-entropy loss function is utilized to suppress the negative influence of sample imbalance on the change detection results. The binary cross-entropy loss is defined as follows:

$$L_{bce} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log p_i + (1 - y_i)\log\left(1 - p_i\right)\right] \tag{6}$$

where N is the total number of pixels; y_i represents the label of pixel i; $y_i = 0$ and $y_i = 1$ represent the unchanged and changed samples, respectively; and p_i is the predicted probability that pixel i will change.

IV. Experiments

This section presents the comparison of the proposed EGASF network with some state-of-the-art methods with four pairwise real datasets to verify its feasibility and superiority. Some experiments were designed to discuss the relationship between the detection accuracies and the parameters of the proposed EGASF network. The details of the comparisons are provided as follows.

The experimental evaluation covers four representative BCD datasets with different geographic regions, spatial resolutions, building distributions, and change scenarios, together with eight state-of-the-art comparison methods representing diverse technical architectures. In addition, quantitative comparisons, visual analyses, ROC and PR curves, areas under the receiver (AUR) and areas under the precision–recall (AUP) evaluations, ablation studies, and training-ratio experiments are conducted to comprehensively evaluate the proposed EGASF network.

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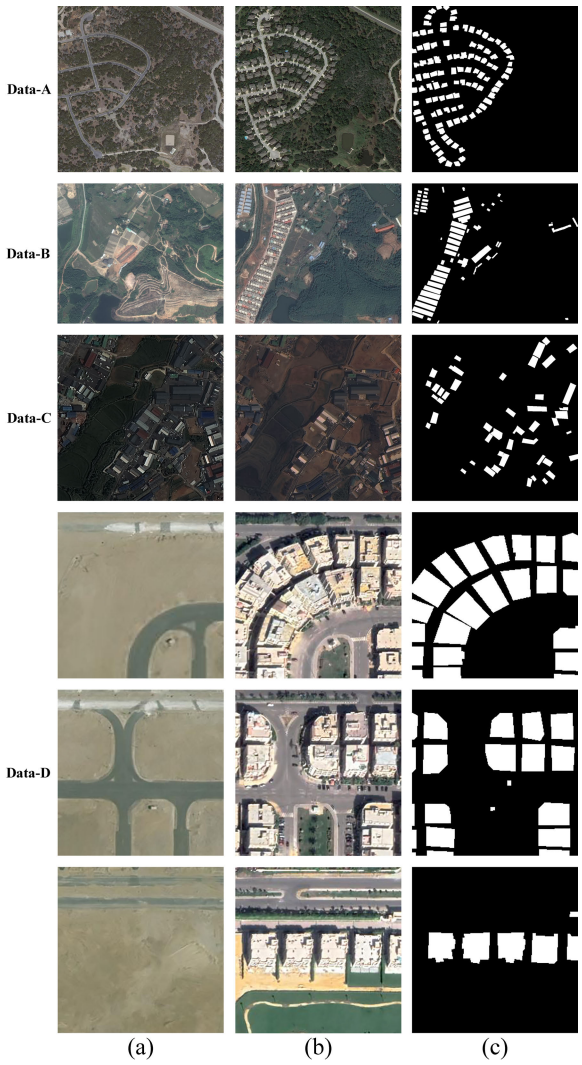


Fig. 5. Datasets A–D. (a) Pre-event image, (b) post-event image, and (c) ground truth.

A. Dataset Description

Four pairs of bitemporal RSIs with very high spatial resolution are adopted to test the performance of the proposed EGASF network (Fig. 5). The details of the datasets are provided as follows.

1) *LEVIR-CD Dataset (Data-A):* The open dataset called LEVIR-CD has been released for BCD in [48]. This dataset consists of 445 training datasets, 128 test datasets, and 64 validation datasets, and a total of 31 333 changed buildings. Each image has a size of 1024 × 1024 pixels with a spatial resolution of 0.5 m/pixel. These images are selected from 20 areas in several cities in Texas, USA.
2) *GZ Dataset (Data-B):* The GZ dataset covers densely built villages in north-central Guangzhou, China [49]. The images have been acquired between 2006 and 2019. These images have a spatial resolution of 0.55 m/pixel, and the size ranges from 1006 × 1168 pixels to 4936 × 5224 pixels.
3) *S2Looking Dataset (Data-C):* The dataset consists of 5000 registered bitemporal image pairs from rural areas around the world with a size of 1024 × 1024 and a spatial resolution of 0.5–0.8 m/pixel, newly built and demolished buildings caused the land-cover change, respectively [50].
4) *EGY-BCD Dataset (Data-D):* The EGY-BCD dataset was proposed for BCD in [51], as shown by some examples in rows four to six of Fig. 5. The dataset was obtained from Google Earth for four urban and coastal areas in Egypt between 2015 and 2022. This dataset contains 6091 pairs of small images of size 256 × 256 × 3, and the spatial resolution of each pair of images is 0.25 m/pixel.

B. Experimental Settings

Eight state-of-the-art methods were selected for comparison to verify the superiority of the proposed approach. The characteristics and parameters of each approach are as follows.

1) *DTCDSCN [27]:* This method is a dual-task constrained deep Siamese convolutional network that can accomplish change detection and semantic segmentation, which can help in learning other discriminative object-level features and obtain a complete change detection map.
2) *LGP-Net [19]:* The method is a local–global pyramid network that combines a local feature pyramid module and a global spatial pyramid module for various building feature extraction.
3) *BIT [39]:* This approach concentrated on the contexts in the spatial–temporal domain through a bitemporal image transformer, allowing the network to explore the high-level discriminative features of the object.
4) *SNU-Net [52]:* The selected SNU-Net addresses the problem of the pixel loss at the edge of the changed target as a result of neglecting the importance of shallow-layer information. SNU-Net alleviates the loss of localization information through compact information transmission between the encoder and the decoder.
5) *PATM-Net [53]:* This approach concentrates on extracting prior-information building and constructing cross-temporal context features. Thereafter, this approach fuses prior extraction and contextual feature by learning a prior-aware transformer to obtain an improved detection map.
6) *ENCL-Net [28]:* This method was proposed for learning the relationship between the semantic label and the reflectance of pixels. Moreover, this mechanism presented a novel edge neighborhood contrastive learning method to force the network to learn discriminative and compact features.
7) *AER-Net [38]:* The network uses a global context feature aggregation module to aggregate information from the extracted multilayer context features to obtain an improved detection map.
8) *USSFC-Net [21]:* The selected USSFC-Net focuses on the cooperation between spatial and spectral features. This method proposes an efficient ultralightweight spatial-spectral feature cooperation network for improving change detection efficiency. A multiscale decoupled

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TABLE I
DETAILED DESCRIPTION OF EACH EVALUATION INDICATOR USED IN QUALITATIVE COMPARISON

Evaluation Indicators	Formula	Explanation
Overall Accuracy (OA)	$OA = \frac{TP+TN}{TP+TN+FP+FN}$	Ratio between the correctly detected pixels in the result and the total number of the ground truth map
Kappa Coefficient (Ka)	$P_0 = \frac{TP+TN}{TP+TN+FP+FN}$ $P_e = \frac{(TP+FP)\times(TP+FN)+(FN+TN)\times(FP+TN)}{(TP+TN+FP+FN)^2}$ $Ka = \frac{P_0-P_e}{1-P_e}$	Ka reflects the reliability of the detection map by measuring the inter-rater reliability for the changed and unchanged items.
Average Accuracy (AA)	AA = $(\frac{TP}{TP+FP} + \frac{TN}{TN+FN})/2$	Average accuracy for the corrected change detection accuracy and unchanged detection accuracy.
False Alarm (FA)	$FA = \frac{FP}{TN+FP}$	Ratio between the number of pixels that are detected as "changed pixels" but are unchanged in the ground truth map and the number of pixels that are unchanged in the ground truth map.
Missing Alarm (MA)	$MA = \frac{FN}{TP+FN}$	Ratio between the number of pixels that are identified as "unchanged" but are changed in the ground truth map and the number of pixels that are changed in the ground truth map.
Total Error (TE)	$TE = \frac{FP+FN}{TP+TN+FP+FN}$	Ratio betwen the number of incorrectly detected pixels in the result and the total number of the ground truth map.
F1-Score	$F1 = 2 \times \frac{precision\times recall}{precision+recall}$	F1 score is an indicator used to measure the accuracy of the dichotomous model in statistics. It considers the accuracy and recall of classification models. The F1 score can be regarded as a harmonic average of model accuracy and recall.

convolution is designed to flexibly capture the multiscale features of changed objects.

The parameters for all the deep learning methods are the same to ensure fairness of experimental comparison (i.e., learning rate = 0.0001, batch size = 256 $\times$ 256, and the ratio between the training and test samples is 2:8). The detailed description of the evaluation metrics is summarized in Table I.

C. Results and Analysis

The quantitative and visual comparisons based on the above-mentioned parameter settings are presented as follows.

The quantitative comparisons between the proposed approach and the eight state-of-the-art methods are summarized in Tables II–V. The results of the quantitative comparisons indicated that the proposed approach achieved the best accuracies across several indicators. Compared with the best-performing competing method for each metric, the proposed EGASF achieves OA, mIoU, and $F1$-score improvements of 0.30, 1.08, and 1.02 percentage points on Data-A, respectively. The corresponding improvements are 0.82, 4.16, and 4.90 percentage points on Data-B. On Data-C, EGASF improves OA and mIoU by 0.52 and 1.49 percentage points, respectively, while its $F1$-score is only 0.04 percentage points lower than the best result. On Data-D, although the OA and mIoU are 0.50 and 0.03 percentage points lower, respectively, EGASF improves AA, CIoU, and $F1$-score by 2.30%, 0.67%, and 0.50% points.

The visual performance comparisons supported the conclusion of the quantitative comparisons. The visual comparisons in Figs. 6–9 indicated that the maps achieved by the approach have few false detection pixels (blue pixels) and missed detection pixels (red pixels). Moreover, the shape and size of the corrected detection buildings are better preserved in the detection maps achieved by the proposed approach compared with those of other approaches.

D. Discussion

Some experiments are designed for discussion to further verify the feasibility of the proposed approach. These experiments include the module's ablation, change magnitude comparisons, and the relationship between detection accuracies and training–testing ratio. Details of the discussion are presented as follows.

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TABLE II
COMPARISON BETWEEN THE PROPOSED AND THE EIGHT STATE-OF-THE-ART RELATED METHODS FOR DATA-A. THE BEST RESULTS ARE GIVEN IN BOLD

Methods	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	Precision	Recall	F1
DTCDSCN[27]	96.39	0.79	84.59	0.35	30.48	3.61	67.60	96.10	81.85	96.06	69.52	80.67
LGP-Net[19]	96.71	0.82	88.13	0.92	22.82	3.29	71.75	96.41	84.08	91.07	77.18	83.55
BIT[39]	96.32	0.80	89.37	1.87	19.38	3.77	69.85	95.87	82.86	83.94	80.62	82.25
SNU-Net[52]	97.24	0.86	92.55	1.46	13.45	2.76	77.27	96.96	87.12	87.82	86.55	87.18
PATM-Net[53]	97.22	0.84	88.76	0.44	22.05	2.78	75.22	96.97	86.09	95.56	77.95	85.86
ENCL-Net[28]	97.41	0.87	94.15	1.69	10.01	2.59	79.00	97.13	88.07	86.62	89.99	88.27
AER-Net[38]	97.61	0.87	92.04	0.86	15.07	2.39	79.34	97.36	88.35	92.34	84.93	88.48
USSFC-Net[21]	97.47	0.87	**94.54**	1.72	**9.20**	2.53	79.55	97.20	88.38	86.53	**90.80**	88.61
Proposed EGASF	**97.91**	**0.88**	91.55	**0.33**	16.57	**2.09**	**81.21**	**97.70**	**89.46**	**96.83**	83.43	**89.63**

TABLE III
COMPARISON BETWEEN THE PROPOSED AND THE EIGHT STATE-OF-THE-ART RELATED METHODS FOR DATA-B. THE BEST RESULTS ARE GIVEN IN BOLD

Method	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	PRE	REC	F1
DTCDSCN[27]	93.95	0.66	86.82	4.54	21.81	6.05	53.07	93.50	73.29	62.29	78.19	69.34
LGP-Net[19]	95.60	0.71	83.10	1.75	32.05	4.40	57.50	95.32	76.41	78.89	67.95	73.01
BIT[39]	94.38	0.68	86.63	3.97	22.78	5.62	54.63	93.98	74.31	65.13	77.22	70.66
SNU-Net[52]	95.39	0.73	88.62	3.17	19.59	4.61	60.45	95.04	77.75	70.88	80.41	75.35
PATM-Net[53]	96.43	0.78	**89.04**	2.00	19.92	3.57	66.26	96.16	81.21	79.34	80.08	79.71
ENCL-Net[28]	94.66	0.56	71.17	0.36	57.31	5.35	41.16	94.45	67.80	91.97	42.69	58.31
AER-Net[38]	93.14	0.64	88.94	5.98	**16.14**	6.87	51.68	92.59	72.14	57.39	**83.86**	68.15
USSFC-Net[21]	96.76	0.76	82.98	**0.32**	33.71	3.25	64.14	96.56	80.35	95.20	66.29	78.16
Proposed EGASF	**97.58**	**0.83**	87.77	0.33	24.14	**2.42**	**73.33**	**97.41**	**85.37**	**95.65**	75.86	**84.61**

TABLE IV
COMPARISON BETWEEN THE PROPOSED AND THE EIGHT STATE-OF-THE-ART RELATED METHODS FOR DATA-C. THE BEST RESULTS ARE GIVEN IN BOLD

Method	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	PRE	REC	F1
DTCDSCN[27]	91.75	0.35	64.85	3.05	67.25	8.25	24.32	91.53	57.93	48.60	32.75	39.13
LGP-Net[19]	90.18	0.39	**70.73**	6.06	52.47	9.82	28.15	89.79	58.97	40.85	47.53	43.94
BIT[39]	91.66	0.46	74.18	4.96	**46.67**	8.34	**34.11**	91.28	62.70	48.63	**53.33**	**50.87**
SNU-Net[52]	90.97	0.40	70.11	5.00	54.78	9.03	28.85	90.63	59.74	44.35	45.22	44.78
PATM-Net[53]	93.94	0.42	64.77	0.42	70.05	6.06	28.58	93.79	61.19	86.22	29.95	44.46
ENCL-Net[28]	91.43	0.38	67.98	4.04	60.01	8.57	27.42	91.14	59.28	46.60	39.99	43.04
AER-Net[38]	93.39	0.44	67.20	1.55	64.05	6.61	30.56	93.19	61.87	67.09	35.95	46.82
USSFC-Net[21]	92.74	0.45	70.20	2.90	56.70	7.26	32.56	92.48	62.52	56.78	43.30	49.13
Proposed EGASF	**94.46**	**0.48**	67.51	**0.33**	64.66	**5.54**	34.07	**94.30**	**64.19**	**90.51**	35.34	50.83

The proposed EGASF network comprises the backbone of UNet and ASFF, MFDM, and BEDE modules for clarity of presentation. In this ablation experiment, each submodule is gradually embedded in the BCD task for Dataset-A. The ablation comparison of the different submodules for Dataset-A demonstrated the feasibility and effective-

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TABLE V
COMPARISON BETWEEN THE PROPOSED AND THE EIGHT STATE-OF-THE-ART RELATED METHODS FOR DATA-D. THE BEST RESULTS ARE GIVEN IN BOLD

Method	OA	Ka	AA	FA	MA	TE	UCIoU	CIoU	mIoU	PRE	REC	F1
DTCDSCN[27]	66.05	0.28	77.55	40.54	**4.36**	33.91	58.88	34.09	46.49	34.63	**95.64**	50.85
LGP-Net[19]	69.41	0.35	74.79	33.70	16.73	30.59	63.90	33.30	48.60	35.68	83.27	49.96
BIT[39]	91.99	0.60	83.54	3.11	29.80	8.01	**90.81**	61.65	76.23	83.51	70.20	76.28
SNU-Net[52]	89.47	0.53	82.94	6.76	27.36	10.53	87.85	55.84	71.84	70.71	72.64	71.66
PATM-Net[53]	**92.06**	0.59	84.28	3.44	27.99	**7.94**	90.85	62.44	76.65	82.46	72.01	76.88
ENCL-Net[28]	90.58	0.51	77.69	1.95	42.67	9.42	89.48	52.75	71.11	86.84	57.33	69.07
AER-Net[38]	68.73	0.33	79.96	37.77	2.31	31.27	61.91	36.42	49.16	36.73	97.69	53.39
USSFC-Net[21]	91.82	0.57	80.30	**1.50**	37.90	8.18	90.77	58.20	74.49	**90.28**	62.10	73.58
Proposed EGASF	91.56	**0.60**	**86.58**	5.56	21.26	8.44	90.14	**63.11**	**76.62**	76.07	78.74	**77.38**

TABLE VI
ABLATION EXPERIMENT COMPARISON BASED ON DATA-A. THE BEST RESULTS ARE GIVEN IN BOLD

Models	Backbone	ASFF	MFDM	BEDE	OA	FA	mIoU	F1
	(M1)	(M2)	(M3)	(M4)				
M1	√				96.86	1.98	85.79	85.75
M1+M2	√	√			97.39	1.59	87.90	88.07
M1+M2+M3	√	√	√		97.49	1.50	88.27	88.47
M1+M2+M3+M4	√	√	√	√	**97.91**	**0.33**	**89.46**	**89.63**

TABLE VII
QUANTITATIVE COMPARISON IN TERMS OF AUR AND AUP FOR CMIS GENERATED BY DIFFERENT METHODS ON DATASET-A–DATASET-D. THE BEST RESULTS ARE GIVEN IN BOLD

Methods	Data-A		Data-B		Data-C		Data-D		Average	
	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP	AUR	AUP
DTCDSCN[27]	0.948	0.913	0.894	0.808	0.729	0.397	0.959	0.929	0.882	0.762
LGP-Net[19]	0.936	0.896	0.878	0.781	0.743	0.450	0.943	0.893	0.875	0.755
BIT[39]	0.931	0.866	0.901	0.801	**0.789**	0.563	0.992	0.988	0.903	0.805
SNU-Net[52]	0.941	0.922	0.899	0.815	0.719	0.487	0.974	0.963	0.883	0.797
PATM-Net[53]	0.954	0.932	0.920	0.858	0.659	0.607	0.986	0.977	0.880	0.844
ENCL-Net[28]	0.966	0.930	0.720	0.703	0.702	0.447	0.867	0.884	0.814	0.741
AER-Net[38]	**0.987**	0.958	0.920	0.835	0.745	0.509	0.965	0.925	0.904	0.807
USSFC-Net[21]	0.959	0.912	0.864	0.852	0.737	0.512	**0.995**	**0.994**	0.889	0.817
Proposed EGASF	0.978	**0.962**	**0.922**	**0.896**	0.770	**0.625**	0.971	0.938	**0.910**	**0.855**

ness of the proposed submodules in the proposed approach (Table VI). For example, the backbone without any submodules obtained the lowest accuracies. When embedding the proposed submodules one-by-one in the backbone, the detection accuracies gradually improved. Meanwhile, the backbone with all the proposed submodules achieved the best accuracies. The ablation observations effectively verified the performance of each proposed submodule in the proposed approach.

Compared with the backbone, introducing the ASFF module improves OA, mIoU, and $F1$-score by 0.53, 2.11, and 2.32 percentage points, respectively, while reducing FA by 0.39 percentage points. This demonstrates the effectiveness of the adjacent-scale fusion mechanism. The MFDM module further improves OA, mIoU, and $F1$-score by 0.10, 0.37, and 0.40 percentage points, respectively. After introducing the BEDE module, OA, mIoU, and $F1$-score are further improved by 0.42, 1.19, and 1.16 percentage points, respectively, while

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TABLE VIII
COMPUTATIONAL COMPLEXITY AND INFERENCE EFFICIENCY COMPARISONS

Method	Params (M)	FLOPs (G)	Inference Time (ms/pair)	FPS
AERNet[38]	25.36	11.77	262.66	3.81
DTCDSCN[27]	41.07	15.19	413.62	2.42
LGPNet[19]	71.34	111.19	1555.21	0.64
Proposed EGASF	34.89	17.00	354.70	2.82

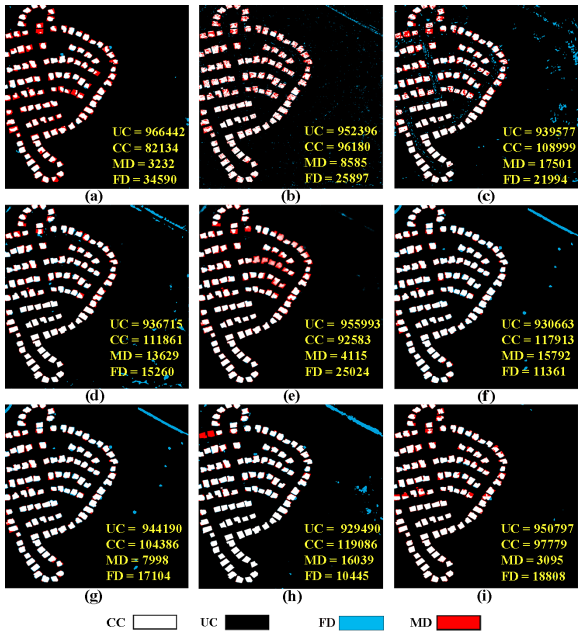


Fig. 6. Detection maps acquired with different methods for Data-A. (a) DTCDSCN [27], (b) LGP-Net [19], (c) BIT [39], (d) SNU-Net [52], (e) PATM-Net [53], (f) ENCL-Net [28], (g) AER-Net [38], (h) USSFC-Net [21], and (i) proposed EGASF.

FA decreases from 1.495% to 0.331%. These results verify the effectiveness of the edge-guided strategy in suppressing false alarms and improving BCD. The visual performance of the CMIs obtained with different approaches is discussed. Here, ROC and PR were adopted for quantitative comparisons. As shown in Fig. 10, our proposed approach achieved better performance in terms of the ROC and PR curves of numerous datasets when compared with those of the selected methods. Apart from the visual observation, quantitative comparisons were also observed in terms of AUR operating characteristics curve and the AUP curve. Our method can acquire better AUR and AUP on these four datasets, and it achieves the best average AUP and AUR compared with the other comparison methods (Table VII). Therefore, our proposed method performs stably on different datasets.

As shown in Table VIII, the proposed EGASF contains 34.89 M parameters and requires 17.00 G FLOPs, with an average inference time of 354.70 ms per bitemporal image pair and a processing speed of 2.82 pairs/s. Compared with DTCDSCN and LGP-Net, EGASF requires fewer parameters and achieves faster inference. Although AER-Net exhibits lower computational complexity and

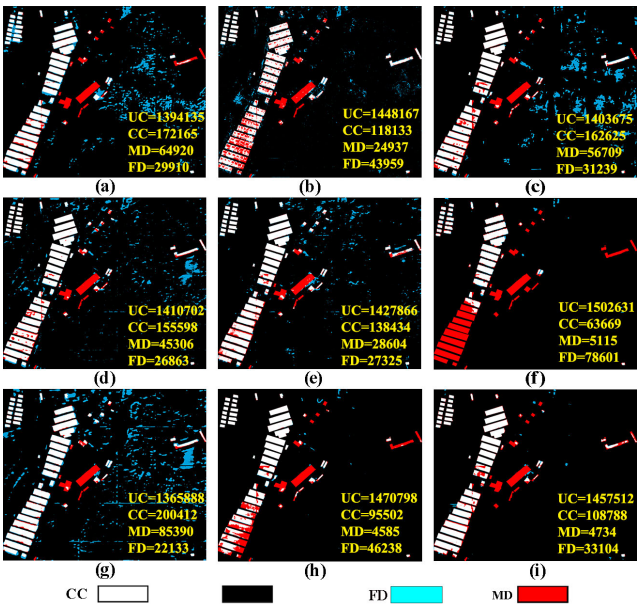


Fig. 7. Detection maps acquired with different methods for Data-B. (a) DTCDSCN [27], (b) LGP-Net [19], (c) BIT [39], (d) SNU-Net [52], (e) PATM-Net [53], (f) ENCL-Net [28], (g) AER-Net [38], (h) USSFC-Net [21], and (i) proposed EGASF.

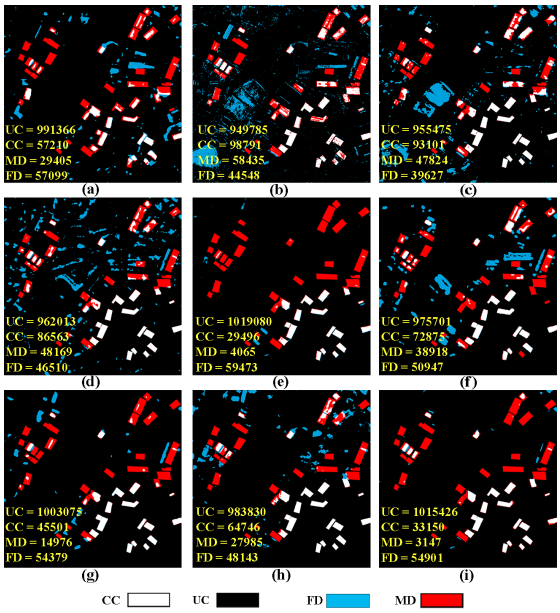


Fig. 8. Detection maps acquired with different methods for Data-C. (a) DTCDSCN [27], (b) LGP-Net [19], (c) BIT [39], (d) SNU-Net [52], (e) PATM-Net [53], (f) ENCL-Net [28], (g) AER-Net [38], (h) USSFC-Net [21], and (i) proposed EGASF.

higher inference efficiency, EGASF achieves better overall detection performance, demonstrating a reasonable trade-off between detection accuracy and computational cost.

For large-scale VHR remote sensing images, the bitemporal image pair can be divided into fixed-size overlapping patches and processed sequentially. The patch-level predictions can then be stitched to generate the complete building change map. This strategy avoids loading the entire large image into

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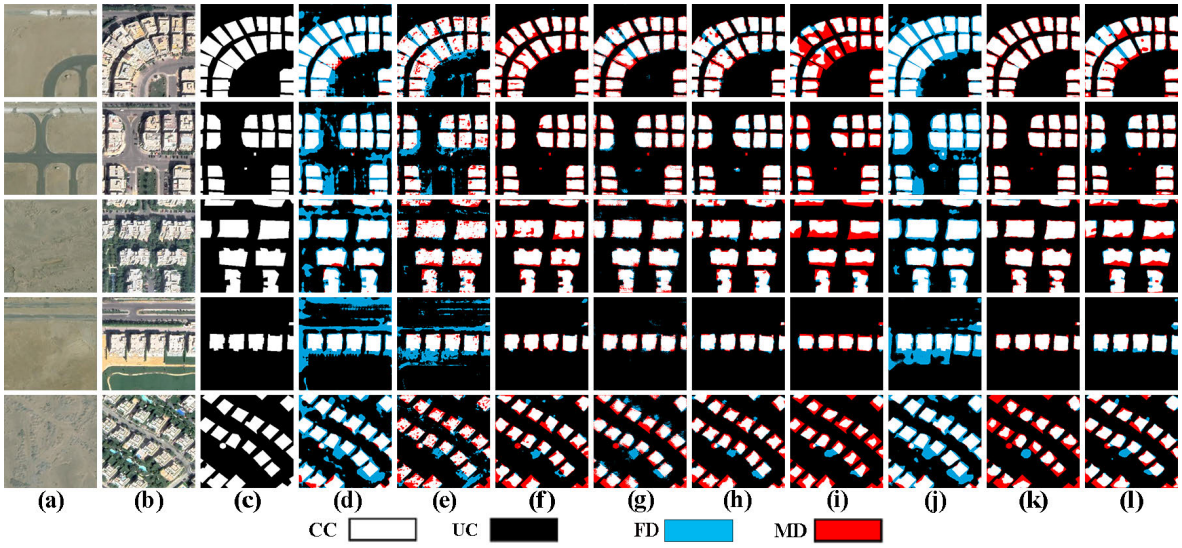


Fig. 9. Some examples of detection maps obtained using different methods on Data-D. (a) Pre-event image, (b) post-event image, (c) ground truth, (d) DTCDSCN [27], (e) LGP-Net [19], (f) BIT [39], (g) SNU-Net [52], (h) PATM-Net [53], (i) ENCL-Net [28], (j) AER-Net [38], (k) USSFC-Net [21], and (l) proposed EGASF.

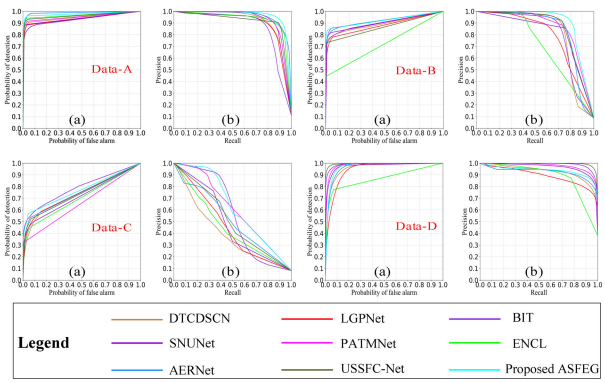


Fig. 10. Comparisons of ROC and PR curves generated by different methods for each dataset. (a) ROC curves and (b) PR curves.

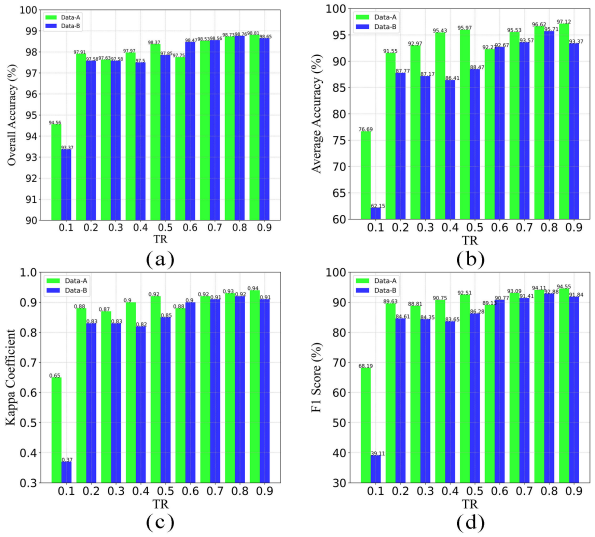


Fig. 11. Accuracy comparisons are obtained by applying the proposed approach with TR, where TR is the ratio between training samples and total samples. (a) Overall Accuracy (OA) versus TR. (b) Average Accuracy (AA) versus TR. (c) Kappa Coefficient (Ka) versus TR. (d) F1-Score versus TR.

GPU memory and enables EGASF to process large-area VHR remote sensing images. Nevertheless, the total processing time increases with the number of patches, and lightweight network design and inference acceleration will be investigated in future work.

The relationship between detection accuracies and the ratio of training and testing samples was discussed to promote the application of this approach in practical scenarios. Fig. 11 demonstrates that the indicators in terms of OA, AA, Ka, and $F1$-score were improved with the increase in the number of training samples. Moreover, the proposed approach achieved lower detection accuracies when the ratio between the training and the testing samples was 0.1. When the ratio between training and testing samples is increased from 0.2 to 0.9, the detection accuracies increased, with slight variations. This result means that the proposed network can obtain a satisfactory BCD map with RSIs when 20% of the samples were used for training.

V. Conclusion

In this article, a novel approach called the EGASF network is proposed for BCD with RSIs. In the proposed EGASF network, the connection feature and difference feature were concentrated to improve the performance of BCD tasks with RSIs. Three submodules, namely, ASFF, MFDM, and BEDE modules, were proposed and composed of the proposed EGASF network. The experiments with four pairwise RSIs and comparison with eight state-of-the-art methods effectively demonstrated the superiority of the proposed EGASF for BCD with RSIs. Moreover, the ablation experiments verified the feasibility of each proposed module in the proposed approach.

Although the effectiveness of the proposed EGASF network has been demonstrated on four real-world datasets, several

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limitations remain. Its robustness under more complex scenarios, such as severe illumination and seasonal variations, registration errors, and densely distributed buildings, requires further investigation. In addition, the current method mainly focuses on bitemporal optical images and does not fully exploit complementary information from multimodal data, such as SAR and LiDAR. Future work will investigate multimodal feature integration, domain adaptation, and unsupervised or self-supervised learning to improve the generalization capability of the model across different regions and imaging conditions.

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